EvidenceLens: An Offline, Explainable Research Search Agent over TREC-COVID Built from Classical Information Retrieval

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Abstract

EvidenceLens is an offline search agent for scientific literature. It searches the 171,332 papers of TREC-COVID without any generative AI at query time. It extends our midsem project Goldilocks, whose Boolean queries came from a language-model API, by replacing every model call with classical retrieval: concept extraction by idf, a Boolean controller that keeps the result set between 20 and 200 papers, a word2vec thesaurus, Rocchio pseudo-relevance feedback, BM25 ranking, clarifying questions built by leader-follower clustering, rule-based follow-up handling, and a rule-based study-type classifier whose label can filter or re-weight results. Every step is logged and shown in a web app. On the 50 TREC-COVID topics the agent is statistically tied with full-collection BM25 on keyword queries (nDCG@10 0.5897 against 0.6032, $p = 0.67$) and clearly worse on natural-language questions (0.4452 against 0.6555, $p = 0.0001$). It buys a reading-sized result set (96–98% of topics inside the band) with a loss in recall that we report. The accuracy of the study-type classifier against hand labels is **TODO(metric)**.

1 Introduction

Background. A researcher who searches a large collection of papers meets two opposite problems: a short query returns thousands of results, and a long, specific query returns a few or none. The COVID-19 literature made this concrete. TREC-COVID [1] built a test collection for it, with 50 topics and graded judgments over an evolving set of papers.

Importance. Ranking alone does not fix this. A ranked list of 10,959 papers (the number our Boolean query for "covid vaccine" matches) leaves the reader to guess what the question really was, and a ranker gives no reason for its order. For scientific work, the study type also matters: a systematic review and a case report are different kinds of evidence.

Motivation. Our midsem project, Goldilocks, controlled result size with a Boolean query controller but depended on a language-model API to propose queries. Free keys were unreliable, and 14 of 50 evaluation topics had no valid proposal. We asked how much of that job classical IR can do offline, with every decision inspectable.

Contributions. We do not claim a new retrieval algorithm. The contributions are an integration and its evaluation:

- An offline agent that replaces all language-model calls of Goldilocks with concept extraction, a thesaurus, relevance feedback and logged rules.
- Clarifying questions from clustering the candidate set, and follow-up handling with five named rules, both shown to the user.
- An evidence-type lens: a rule-based classifier with an evidence score g(d) taken from a published level-of-evidence ordering, used as filter and optional score.
- A paired-randomization evaluation of six runs on both topic fields, with the losses stated.
- A web app whose default view shows results only; the reasoning opens on request.

2 Literature Review / Related Work

Test collection. Roberts et al. [1] describe TREC-COVID, a shared task run in five rounds from April to July 2020 with 50 topics. BEIR [2] includes it as one of its zero-shot benchmark datasets; the paper reports BM25 with nDCG@10 = 0.656 on it (Table 2, version v4), a strong baseline we compare against. Our difference from both: we use the collection as the corpus of an interactive agent and evaluate the agent against that baseline, not only ranking models.

BM25. Robertson and Zaragoza [3] present the probabilistic relevance framework behind BM25. We implement BM25 ourselves and use it as the ranker and as the main baseline; we do not extend the model.

Clarifying questions. Aliannejadi et al. [6] formulate asking clarifying questions in open-domain conversations, build the Qulac dataset, and propose a framework with question retrieval, question selection and document retrieval. Zamani et al. [7] generate clarifying questions for web queries, starting from a taxonomy derived from Bing query reformulations and using tem-

plates plus supervised and reinforcement learning models. Both rely on data we do not have for scientific search. We instead cluster the top 300 candidates of the current result set and label each cluster with its most distinctive terms, so the "question" is a choice between topics found in the data. This is simpler and unlearned, and we have no human study showing the choices help.

Conversational search. Radlinski and Craswell [8] propose a theoretical framework for conversational search. We do not implement that framework. Our conversational layer is a small set of rules for follow-ups, filters and exclusions, each visible to the user.

Levels of evidence. Burns, Rohrich and Chung [9] describe levels of evidence for evidence-based medicine. We use their ordering to rank our study-type classes (Section 5.7); the numeric g(d) values are our own choice.

3 Objective

Build a search agent over a fixed scientific collection that (1) needs no generative AI at query time, (2) returns result sets of a size a person can read, (3) asks for clarification when a question is too broad and carries context across turns, (4) labels each paper with its study type, and (5) explains every step. Then measure what this costs against plain BM25.

4 Proposed Model

Figure 1 shows the architecture. A question is turned into concepts (stemmed words or groups of synonyms from an 18-group COVID lexicon). The controller builds a Boolean query over the inverted index, counts hits, and tightens or relaxes it until the count is inside the band of 20 to 200 papers. The accepted set is ranked with BM25. If the set is still too large and no move applies, the clarifier clusters it and asks the user to choose. The conversation manager rewrites follow-up turns into a new concept list before the controller runs. The explainer reports, per paper, the matched terms, their BM25 contributions, the study-type rule and the best sentence of the abstract.

5 Methodology

Notation: $N = 171{,}332$ documents, df_t the document frequency of stem t, $\mathrm{tf}_{t,d}$ the term count, L_d the number of indexed tokens of d, and $\overline{L} = 169.52$ the mean length. The textbook [10] is the source of the general techniques below; the code is ours or Raghav Garg's.

5.1 Boolean retrieval

The index stores sorted postings per stem. AND queries are answered by intersecting postings lists in increasing order of df, so the shortest list drives the work. OR is a merge and NOT removes documents. Wildcards use a trigram k-gram index, and `title:t` restricts a term to a title-zone postings file, a weighted-zone idea from the lectures. The Boolean parser handles AND, OR, NOT and parentheses; quoted words are required.

5.2 Ranking

BM25. With the Lucene form of idf,

$$\mathrm{idf}(t) = \ln\Big(1 + \frac{N - \mathrm{df}_t + 0.5}{\mathrm{df}_t + 0.5}\Big), \tag{1}$$

$$\mathrm{bm25}(d,q) = \sum_{t \in q} \mathrm{idf}(t)\, \frac{\mathrm{tf}_{t,d}\,(k_1+1)}{\mathrm{tf}_{t,d} + k_1\big(1 - b + b\,L_d/\overline{L}\big)}, \tag{2}$$

with $k_1 = 1.2$ and $b = 0.75$, not tuned on the test topics. A repeated query term counts once.

ltc.ltc cosine. Document and query weights are $w_{t,x} = (1 + \log_{10} \mathrm{tf}_{t,x}) \cdot \log_{10}(N/\mathrm{df}_t)$ and the score is the cosine of the two weight vectors. (The brief first called this lnc.ltc; the code applies idf on both sides, so ltc.ltc is the correct name.)

5.3 Query expansion and feedback

Thesaurus. word2vec skip-gram [5] is trained on the stemmed title and abstract of every paper (vector size 100, window 5, min count 5, 5 epochs, seed 42). For every stem with df ≥ 5 we keep up to 5 neighbours with cosine ≥ 0.6. Training uses 4 workers, so it is not bit-reproducible.

Rocchio. Rocchio relevance feedback [4] moves the query towards the mean of documents assumed relevant. We use the positive part only ($\gamma = 0$): the top 10 BM25 documents are D_r, a candidate term gets weight $\sum_{d \in D_r} (1 + \log_{10} \mathrm{tf}_{t,d})\, \mathrm{idf}_t$, and the original terms keep weight $\alpha = 1$. In the controller, the best feedback term that keeps at least 20 hits is AND-ed into the query; in run R3, the top 10 terms are added with weight $\beta = 0.5$.

5.4 Controller

Each concept is an OR-group of stems; its document frequency is the size of the union of its postings, so a group of synonyms is as rare as the papers that contain any member. Concepts whose union covers more than 10% of the collection are marked eliminated (index elimination). The first Boolean query ANDs the 3 non-eliminated concepts with the highest idf plus every required (quoted) concept. Then, in each step, the controller runs the query and compares the hit count with the band $[20, 200]$:

1. In band: ACCEPT.
2. Too many hits, first move that applies and keeps ≥ 20 hits: add the next held concept (then eliminated common concepts), require the best stem of the highest-idf concept in the title, AND a Rocchio term. If none applies: CLARIFY.
3. Too few hits: add up to 3 thesaurus neighbours to the rarest concept, then up to 5 k-gram prefix expansions of its rarest stem, then drop the rarest non-required concept. If none applies: EXHAUSTED.

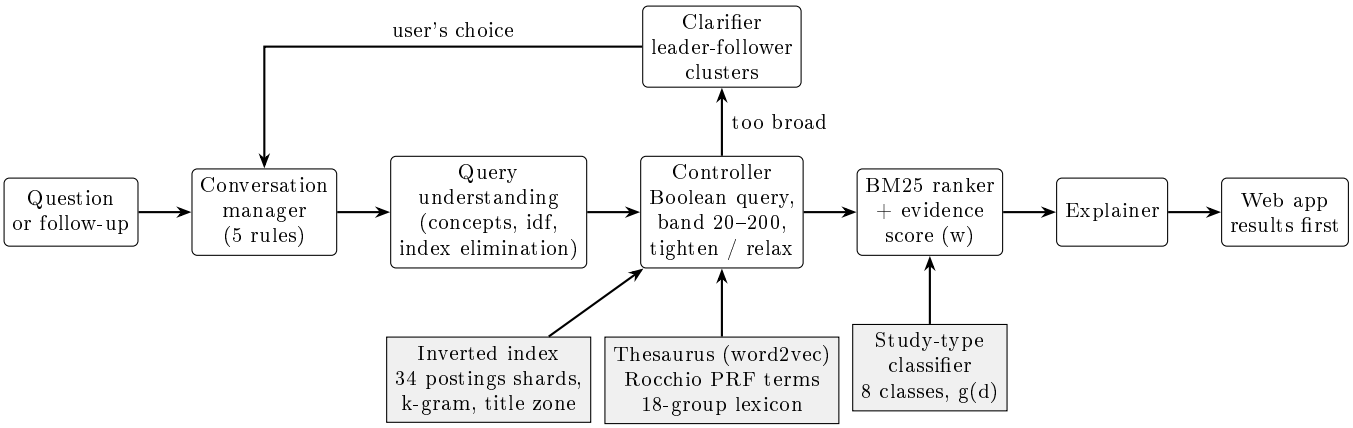


Figure 1: EvidenceLens architecture. Solid boxes are processing steps; shaded boxes are data built offline. No component calls a generative model.

A move is never applied twice to the same concept, and the iteration limit is 8. Required words are never dropped. Under an evidence filter no concept is dropped. The final result set is ranked by BM25. If the set is empty, BM25 over the whole collection is used.

5.5 Clarification by clustering

For a result set still above the band, the top 300 documents by BM25 on the concept stems are candidates, each an ltc vector. With n candidates, $\lceil\sqrt{n}\rceil$ leaders are drawn with a fixed seed (42); every candidate joins the leader with the highest cosine similarity (the cluster-pruning idea of the lectures, used here to group). Clusters under 10 members are dropped, at most the 4 largest are shown, and fewer than 2 means no question. The label of a cluster C is the three stems with the highest contrast score

$$\text{score}(t) = \tfrac{1}{|C|}\sum_{d\in C} w_{t,d} - \tfrac{1}{|D|}\sum_{d\in D} w_{t,d}, \qquad (3)$$

where D is the candidate set. Contrasting with the candidates instead of the whole corpus makes sibling clusters differ from each other; this was our choice after the plan was written and is not compared with the alternative.

5.6 Follow-up rules

Five named rules run before the controller, and each one that fires is shown to the user. FOLLOWUP_DETECT: a leading cue ("what about", "only") or a pronoun makes the turn a follow-up that is added to the session's concepts. EVIDENCE_FILTER: "only reviews/trials/case reports..." sets a study-type filter. EXCLUSION: "without/except X" turns X into `AND NOT`. FACET_SWAP: a new concept whose lexicon facet (population, intervention, outcome) equals one already in the session replaces it. TOPIC_SWITCH: a turn with no cue or pronoun, at least 2 concepts and Jaccard overlap $J = |A \cap B|/|A \cup B| < 0.2$ with the session concepts starts a new topic.

5.7 Evidence rules and net score

The classifier assigns one of eight classes with 14 ordered regular expressions on the lowercased title and abstract; the first match wins, title rules come first, and reviews are tested before trials because reviews of trials mention "randomized". No match gives `other`. g(d) is the value of the class: systematic review or meta-analysis 1.0 (level 1A in [9]), randomized trial 0.85 (1B), observational 0.6 (2B and 3B), case report or series 0.4 (level 4), lab or animal work 0.15 and opinion 0.15 (both level 5), and modelling 0.5 and other 0.5 as neutral values because the source does not rank them. The levels come from the source; the numeric values are ours, and g(d) describes study design, not the quality of a paper. The net score over the top 100 candidates is

$$\text{net}(d) = \frac{\text{bm25}(d)}{\max_{d'} \text{bm25}(d')} + w\, g(d), \qquad (4)$$

and $w = 0$ leaves the BM25 order unchanged. With a filter, the Boolean result is intersected with the documents of that class before each band check.

6 Experimentation and Results

6.1 Setup

Data. TREC-COVID through BEIR: 171,332 documents (title and abstract), 50 topics, graded judgments; 24.6% of documents have no abstract. Both topic fields are used: `metadata.query` (short keywords, "keyword") and `text` (a natural-language question). Relevant means grade above 0. Depth is 1000. nDCG uses linear gain, as trec_eval and BEIR do, so it is not comparable with our midsem numbers (which used $2^g - 1$). The topics average 493.5 judged-relevant documents.

Metrics. Ranked runs: precision at 10 (P@10), nDCG@10, MRR@10, MAP@100 and recall at 100 and at 1000. Set runs (R0, R4, R5), which return one set and no ranking: set precision (the share of the set that is relevant), set recall (the share of all relevant documents that the set contains), their harmonic mean F1, the mean number of hits and the in-band rate.

Table 1: Keyword query (`metadata.query`), mean over 50 topics.

run	P@10	nDCG@10	MRR@10	MAP@100	R@100	R@1000
R0	0.5200	0.4663	0.7397	0.0593	0.0873	0.2108
R1	0.6660	0.6032	0.8287	0.0859	0.1176	0.4161
R2	0.3660	0.3343	0.5340	0.0369	0.0656	0.3538
R3	0.6380	0.5741	0.7822	0.0813	0.1158	0.4488
R4	0.6480	0.5897	0.7857	0.0721	0.0990	0.1217
R5	0.6460	0.5856	0.7619	0.0725	0.0996	0.1217

Table 2: Question text (`text`), mean over 50 topics.

run	P@10	nDCG@10	MRR@10	MAP@100	R@100	R@1000
R0	0.1360	0.1284	0.2695	0.0065	0.0081	0.0091
R1	0.6860	0.6555	0.8875	0.0904	0.1284	0.4543
R2	0.3720	0.3392	0.5454	0.0353	0.0689	0.3681
R3	0.6680	0.6180	0.8037	0.0871	0.1269	0.4824
R4	0.4940	0.4452	0.6929	0.0280	0.0452	0.0531
R5	0.4800	0.4363	0.6836	0.0278	0.0450	0.0531

Runs. R0: Goldilocks offline (AND of keywords, tf-idf). R1: BM25 over the whole collection. R2: ltc.ltc cosine over the whole collection. R3: BM25 plus Rocchio re-ranking. R4: the EvidenceLens agent (controller v2.1, thesaurus, BM25 on the accepted set). R5: R4 with evidence weight $w = 0.2$, a value chosen before any R5 result existed. Nothing else was tuned. Run files and all numbers are saved in `evaluation/`.

6.2 Ranking quality

What these show. BM25 (R1) is the strongest run on both fields. The tf-idf cosine (R2) is far below it. Rocchio re-ranking (R3) did not help: its nDCG@10 is below plain BM25 on both fields (0.5741 against 0.6032, and 0.6180 against 0.6555). The agent (R4) is close to BM25 on keywords and far below it on questions (Figure 2). The agent beats the Goldilocks baseline R0 on both fields.

The recall cap. R4 and R5 return one accepted set, and the band caps it at about 200 documents (mean 128 hits on the question text), while a topic has 493.5 relevant documents on average (minimum 111, maximum 1266). Their Recall@1000 cannot approach that of a ranking run: 0.1217 and 0.0531 for R4 against 0.4161 and 0.4543 for R1. The agent trades recall for a reading-sized set.

6.3 Set size

The controller keeps 98% (keyword) and 96% (question) of topics inside the band, where the midsem AND baseline manages 32% and 12%. This is the gain the agent was built for, and it is paid for in recall.

6.4 Significance

All tests are paired randomization tests over the 50 topics (10,000 sign flips, two-sided). With 20 tests the Bonferroni threshold is $0.05/20 = 0.0025$. The 20 tests are correlated, so the correction is conservative. Table 4

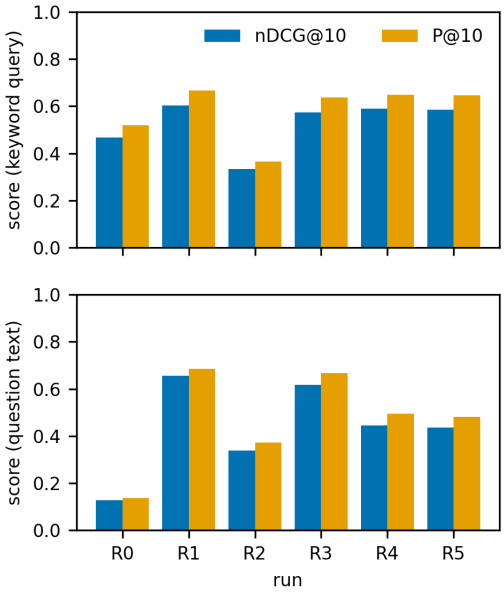


Figure 2: P@10 and nDCG@10 per run, keyword query (top) and question text (bottom).

Table 3: Set metrics for the set runs. R5 returns the same sets as R4.

field	run	hits	prec.	recall	F1	in band
keyword	R0	517.5	0.3324	0.2224	0.2116	32%
keyword	R4	107.4	0.4656	0.1217	0.1732	98%
question	R0	10.1	0.1627	0.0091	0.0145	12%
question	R4	128.2	0.2639	0.0535	0.0776	96%

gives nDCG@10; P@10 leads to the same significant or not-significant conclusion for every pair. **Losses, stated plainly.** (1) On keyword queries the agent is statistically tied with BM25 ($p = 0.6737$), not better. (2) On natural-language questions it is clearly worse ($p = 0.0001$, BM25 wins 38 topics, the agent 12). (3) Pseudo-relevance feedback did not improve BM25; on questions it is nominally lower and significant at 0.05 but not after Bonferroni ($p = 0.0351$). (4) Adding the evidence weight ($w = 0.2$) changed nDCG@10 by 0.0042 and 0.0089, not significant; we make no claim that it improves relevance, and the filter is the feature we rely on. (5) The recall cap above.

On the question field the agent's worst topics show why: for topic 48 the final query requires a group around "midst" and the word "open" and returns nDCG@10 0.000 against BM25's 0.779; for topic 50 it keeps "mrna AND known AND vaccin" and also scores 0.000 against 0.748. Our idf-based concept choice cannot tell the generic words of a long question from its content.

6.5 Comparison with BEIR

BEIR [2], Table 2 of version v4, reports nDCG@10 = 0.656 for BM25 on TREC-COVID. Our R1 on the question field is 0.6555, a difference of −0.0005. We do not claim to reproduce BEIR's pipeline. BEIR's BM25 is Anserini with $k_1 = 0.9$, $b = 0.4$ and separate title and text fields; ours uses $k_1 = 1.2$, $b = 0.75$, one stream of title and abstract tokens and a Lucene-style idf. The

Table 4: nDCG@10 differences (first minus second); wins are topics won by first/second/tied.

field	pair	diff	p	wins	sig.
keyword	R1–R2	+0.2688	0.0001	42/6/2	yes
keyword	R1–R3	+0.0290	0.1279	25/21/4	no
keyword	R1–R4	+0.0134	0.6737	23/23/4	no
keyword	R4–R5	+0.0042	0.6034	15/16/19	no
keyword	R0–R4	−0.1234	0.0019	16/32/2	yes
question	R1–R2	+0.3163	0.0001	43/7/0	yes
question	R1–R3	+0.0375	0.0351	29/18/3	no
question	R1–R4	+0.2103	0.0001	38/12/0	yes
question	R4–R5	+0.0089	0.1418	16/13/21	no
question	R0–R4	−0.3168	0.0001	2/44/4	yes

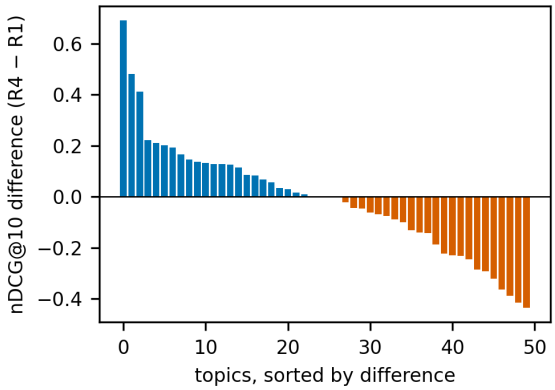


Figure 3: Per-topic nDCG@10 difference, R4 minus R1.

Table 5: Controller versions. v2.1 was designed after seeing v2.0 on all 50 topics.

field	version	in band	nDCG@10	P@10
keyword	v2.0	100%	0.4840	0.5240
keyword	v2.1	98%	0.5897	0.6480
question	v2.0	82%	0.4174	0.4600
question	v2.1	96%	0.4452	0.4940

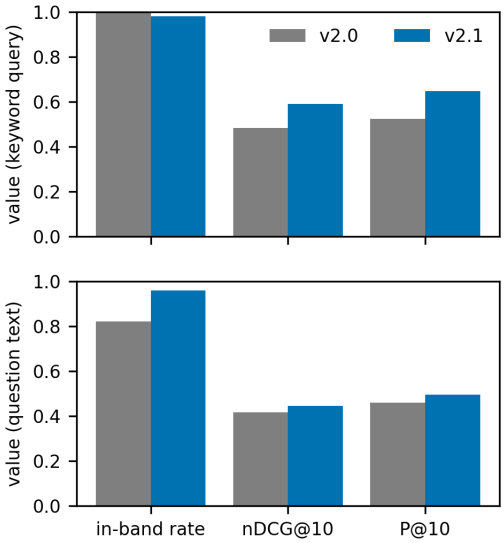


Figure 4: Controller v2.0 and v2.1 (post hoc).

tokenizer and stemming also differ. None of these differences was tested one at a time, so the near match may partly be a coincidence. The BEIR number was read from the version-4 PDF only; earlier versions were not read.

6.6 Controller v2.0 and v2.1 (post hoc)

We revised the tightening order after seeing the v2.0 results on all 50 topics: v2.1 starts with the three highest-idf concepts (the others are held) and tightens with the user's own words before feedback terms. There are no held-out topics, so the v2.1 improvement is a post-hoc result and is probably optimistic. The same applies to two smaller changes after the first controller run: eliminated concepts still count in the BM25 ranking, and no concept is dropped under an evidence filter. The 0.10 elimination threshold was not tuned.

6.7 Case studies

These are scripted case studies, not a benchmark (`evaluation/conversation_runs.json`).

A conversation. "covid vaccine pregnant" starts at 239 hits after the first tightening step and is accepted at 58. The follow-up "what about breastfeeding" fires FOLLOWUP_DETECT (leading cue "what about"), is rewritten to `breastfeeding AND pregnancy AND vaccine AND covid19`, begins with 3 hits, relaxes once with thesaurus neighbours and is accepted at 31. The third turn, "only reviews", fires EVIDENCE_FILTER and FOLLOWUP_DETECT. With the filter the set is empty at first, and after three synonym expansions and two k-gram expansions the controller accepts a set of one document, which is a conference-abstract collection ("XXIV World Allergy Congress 2015"). The filter worked as specified, and the result is a weak answer; we show it because the trace makes the weakness visible.

A clarification. "masks" has one weak concept: 2,658 hits and no tightening move applies, so the controller returns CLARIFY. The four clusters have labels (decontamination, reuse, fit; 64 papers), (surgical, health, use; 63), (airborne, hcws, droplets; 21) and (rcts, randomized, trials; 19). Choosing the first gives the query `(decontamin OR reus OR fit) AND (mask OR respir OR n95) AND title:reus`, which goes from 347 hits to an accepted set of 39.

6.8 The study-type classifier

Class distribution. Over all documents the classifier assigns: other 71.5%, observational 8.7%, lab or animal 8.2%, systematic review 3.0%, modelling 2.7%, case report 2.3%, opinion 2.0%, randomized trial 1.6%. The large `other` class is the default when no rule matches.

Human-validated accuracy: TODO(metric). A blind sample of 120 documents with abstracts is ready for hand labelling (`evaluation/evidence_label_sample.csv`), and nobody has labelled it yet.

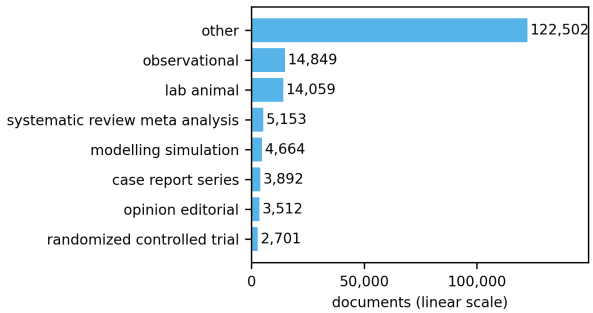


Figure 5: Documents per study-type class over the whole collection.

6.9 Latency

Over 200 in-process requests (the TestClient, no network; 50 topics on both fields and both rankers, one warm-up request first) the saved measurement (`evaluation/api_latency.json`) gives a mean of 192 ms, a median of 128 ms, a 95th percentile of 475 ms and a worst case of 2,014 ms (topic 3, "coronavirus immunity", 1,980 ms of it in the controller). The explanation step takes 26 ms on average. By ranker and field, the mean is 349 ms (agent, keyword), 143 ms (agent, question), 93 ms (BM25, keyword) and 184 ms (BM25, question). The numbers are from one machine and one run.

6.10 The interface

Figure 6 shows the web app. The default Search view shows only a line ("I searched for ... · N papers") and a column of result slips. "How did I get this?" opens a side sheet with the concepts, the numbered trace of tighten and relax steps against the band, the Boolean query and earlier turns, and a switch that re-ranks the current turn with BM25 over the whole collection. This default came from user feedback: the first build showed everything at once and overwhelmed the reader. We have not run a user study of the interface.

7 Conclusions and Limitations

EvidenceLens shows that the query-writing jobs of Goldilocks can be done offline by inspectable classical IR. The agent keeps 96–98% of topics inside a result-size band, is tied with full-collection BM25 on keyword queries, and gives a trace, clarification choices, follow-up rules and a study-type lens that BM25 does not. It does this at a cost: on natural-language questions it is clearly worse than BM25 (nDCG@10 0.4452 against 0.6555, $p = 0.0001$), its recall is capped by the band, and feedback and the evidence weight did not improve ranking. The web app therefore lets the user switch to BM25 over the whole collection.

Limitations. (1) The controller revision v2.1 and two smaller rule changes were made after seeing results, with no held-out topics. (2) The accuracy of the study-type classifier against hand labels is **TODO(metric)**. (3) The classifier is rules on text; g(d) says nothing about the quality of a single paper, and its numeric values are our choice. (4) A filter can leave one weak result (the "only reviews" case study). (5) The thesaurus rebuild is not bit-reproducible. (6) Clarification choices have no user study. (7) One collection, 50 topics, one machine for latency. (8) Cold shard loading still makes the first filtered turn slower than later ones.

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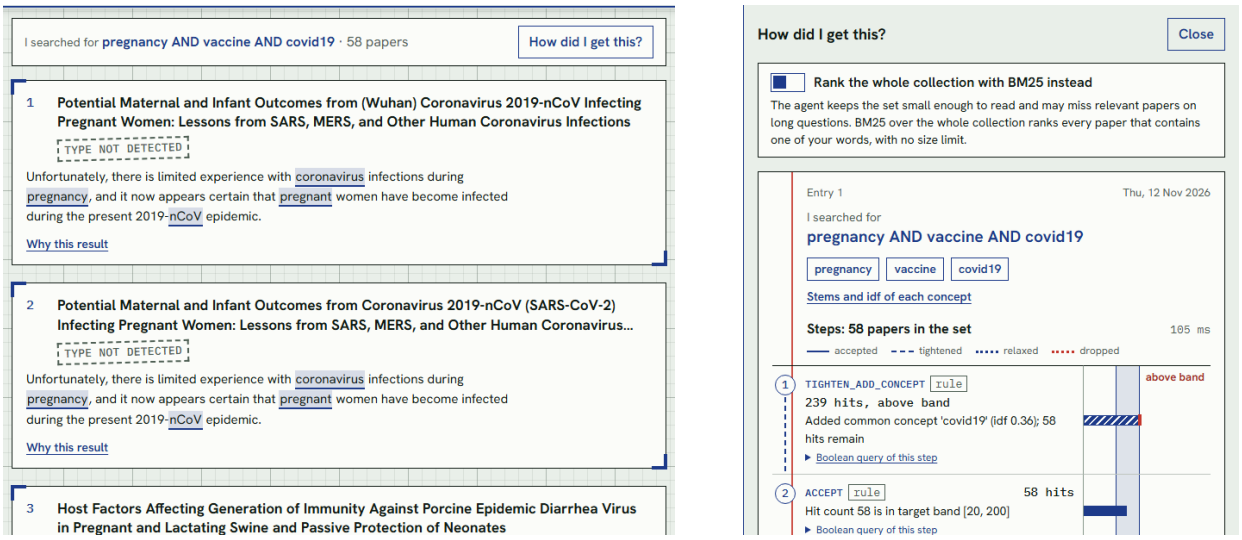


Figure 6: Left: the Search view, results only, cropped from a 1280×720 screen (one line, then the first result slips). Right: the "How did I get this?" sheet open over the results, with the BM25 switch and the start of the notebook entry (concepts, then the numbered steps against the band).

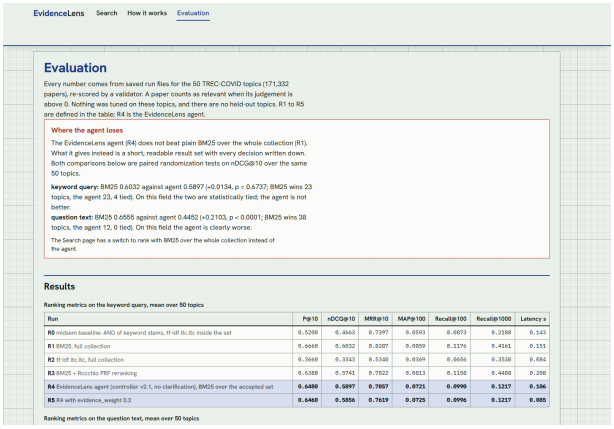


Figure 7: The Evaluation page (layout only; the numbers are in the tables above).